DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
AARHUS UNIVERSITY

Kjeld Laursen
Date:
26 October 2026

Crosstalk

Electronic Hardware System Design – Laboratory Assignment 1

Introduction

This document describes the hands-on group laboratory assignment about crosstalk, which is part of the course Electronic Hardware System Design.

The assignment is mandatory and a report must be written and handed in by the group.
Approval of this report and Laboratory Assignment 2, is required to be allowed to go to the exam!
(An assignment will be set up in BrightSpace for this purpose, where you upload per group.)
(Laboratory Assignment 2 comes later.)

Where

The laboratory work is performed in Circuit-Lab (5126-219).

Deadline

The deadline for hand-in can be seen on BrightSpace.
Note: the assignment is only visible if you are in a group.

HW Test Board Info

The HW test board[1] is shown in Figure 1 and the associated PCB stack-up in Figure 2, is used throughout this assignment.



Figure 1. HW test board for the assignment.

Layer	Type	Cross Section [um]	Cross Section [mil]	Cu (finished, inc. Plating) [um]	Track Width [um]	Track-track spacing [um]
		10	0.4			
1				35	340	200
	Prepreg	180	7.1			
2				35		
	Laminate	200	7.9			
3				35		
	Prepreg	180	7.1			
4				35	217	200
	Laminate	200	7.9			
5				35	217	200
	Prepreg	180	7.1			
6				35		
	Laminate	200	7.9			
7				35		
	Prepreg	180	7.1			
8				35	340	200
		10	0.4			

Figure 2. HW test board PCB stack-up.

[1] Test board is designed and developed by DTU – Electro.

Files for the board

On Brightspace you will find the following files:

- **HW testboard picture DTU.pdf**; A picture of the HW test board.
- **HW testboard schematic DTU.pdf**; A detailed schematic of the HW test board.
- **sn74alvc244-Datasheet.pdf**; Datasheet of the driver IC's.
- **sn74alvc244.ibis.txt**; The ibis file of the driver IC that details the parasitics of the packages.
- **cdcvf2310-Datasheet.pdf**; Datasheet of the clock buffer.

Using the board

The HW test board is powered by USB (use an oscilloscope USB port). The Supply voltage can be adjusted from $2.4\ V$ to $3.4\ V$ using the small potentiometer marked "VCC Adj. 2.4-3.4V". The supply voltage can be measured on probe point P45 marked "VCC" near the middle of the board.

The HW test board includes SN74ALVC244 octal buffers in both the QFN package and the SOIC package. The different outputs of the buffers Q1 (QFN) and Q3 (SOIC) are enabled using the jumpers on the left side of the board.

The input signal for the buffers is supplied by a **20 MHz clock oscillator**. A clock fanout chip (F1) distributes the clock signal and is enabled by two jumpers on the left side of the board.

Carefully study the schematic and component datasheets to understand what each jumper controls and enables.

For each measurement that you make, only enable the outputs that you need, or else you will add ground bounce noise to your crosstalk measurement. (Ground and supply bounce will be explored in laboratory assignment 2)

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Transmission Line Cases

In this assignment we investigate the effect of crosstalk on 3 different transmission line cases, for which we have been given the following capacitance and inductance matrices:

1. **Microstrip line:**

$$C_{matrix} = \begin{bmatrix} 128.5 & -12.0 \\ -12.0 & 128.5 \end{bmatrix} \left[\frac{pF}{m}\right]$$

$$L_{matrix} = \begin{bmatrix} 283.4 & 52.2 \\ 52.2 & 283.4 \end{bmatrix} \left[\frac{nH}{m}\right]$$



2. **Side-by-side stripline:**

$$C_{matrix} = \begin{bmatrix} 153.3 & -21.8 \\ -21.8 & 153.3 \end{bmatrix} \left[\frac{pF}{m}\right]$$

$$L_{matrix} = \begin{bmatrix} 318.5 & 45.3 \\ 45.3 & 318.5 \end{bmatrix} \left[\frac{nH}{m}\right]$$



3. **Broadside stripline:**

$$C_{matrix} = \begin{bmatrix} 167.5 & -53.5 \\ -53.5 & 167.5 \end{bmatrix} \left[\frac{pF}{m}\right]$$

$$L_{matrix} = \begin{bmatrix} 318.0 & 101.4 \\ 101.4 & 318.0 \end{bmatrix} \left[\frac{nH}{m}\right]$$



(Note, that real mutual capacitances are not negative. See lecture slides with regards to the topic.)

The transmission line pairs for the 3 cases are all found at the top of the board, see Figure 3, where one line is driven (aggressor), and the adjacent line is quiet (victim). Study the schematic for details about termination of the lines (or lack thereof).

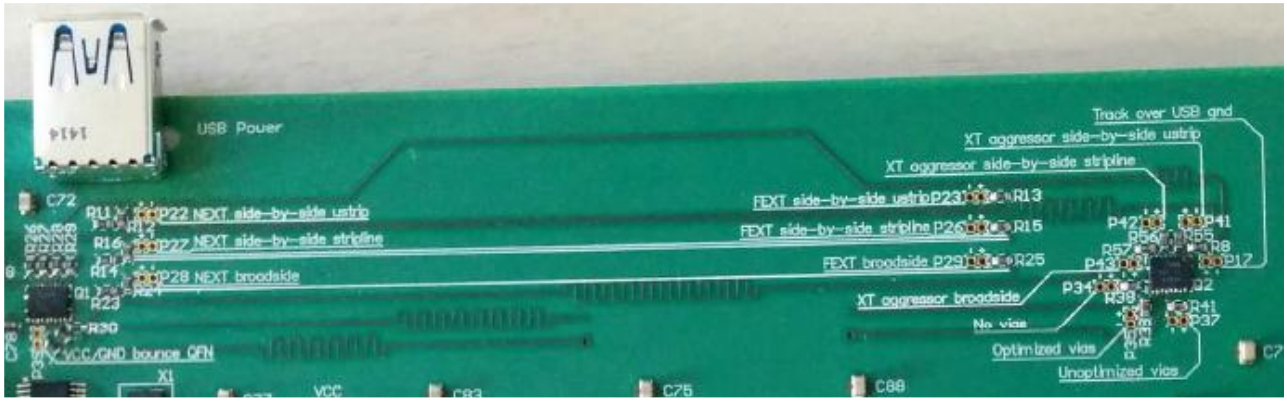


Figure 3. HW test board, focused on crosstalk transmission lines.

The transmission lines are $10\ cm$ long where crosstalk can occur, use that for your models.

For analysis and simulation purposes, assume an edge rate of $500\ ps$, and a nominal supply voltage of $3.3\ V$. (Later, you can adjust the edge rate to more closely match reality if you wish.)

Part 1 – Modelling setup

To perform this part will require studying the HW test board schematic, datasheets, and any other relevant information given about the board. From this a theoretical model of the response at both ends of the driver line and quiet line is to be produced.

Do the following for all three transmission line cases: microstrip line, side-by-side stripline, broad-side stripline.

Task 1.1 Draw models

Draw the equivalent circuit model to be used to simulate the crosstalk in each of the 3 cases.

The model must include the driver circuit (linear model), package parasitics (ibis file), transmission lines (matrices), any relevant board components (board schematic), and receiver circuit (simple model).

Task 1.2 Estimate parameters

Find and estimate the values of the different parameters and components based on data from the HW test-board, board schematic, datasheets, and IBIS model for the driver/receiver IC.

Assume that the aggressor line is matched at the driver end (i.e. output impedance of the drive plus board-level series resistor is designed to match the characteristics impedance of the transmission line).

Task 1.3 Transmission line characteristics

For the transmission lines, calculate the characteristic impedance Z_0, time delay T_D, and propagation velocity v.

Task 1.4 RLCG-model

For the transmission lines calculate the equivalent circuit model of the system (lossless RLCG model) of the coupled transmission lines. This is needed later for the simulation part.

Task 1.5 Noise margins

For data transfers to succeed the signal voltage levels must be within the valid ranges for the receiving buffer. That not only means ensuring that the driver's output is within the acceptable range of the input but also leaving room for a safety margin to increase tolerance to noise.

From the datasheet of the sn74alvc244 buffer find the following:

- Minimum output high level, V_{OH}
- Maximum output low level, V_{OL}
- Minimum input high level, V_{IH}
- Maximum input low level, V_{IL}

(Hint: assume 3.3 V supply and max current load of $\pm 12\ mA$ for both output high and low level.)

From these values calculate the high- and low-level noise margins:

- High level noise margin: $V_{MH} = V_{OH} - V_{IH}$
- Low level noise margin: $V_{ML} = V_{IL} - V_{OL}$

Figure 4 illustrates the different voltage limits and their relation to noise margins.

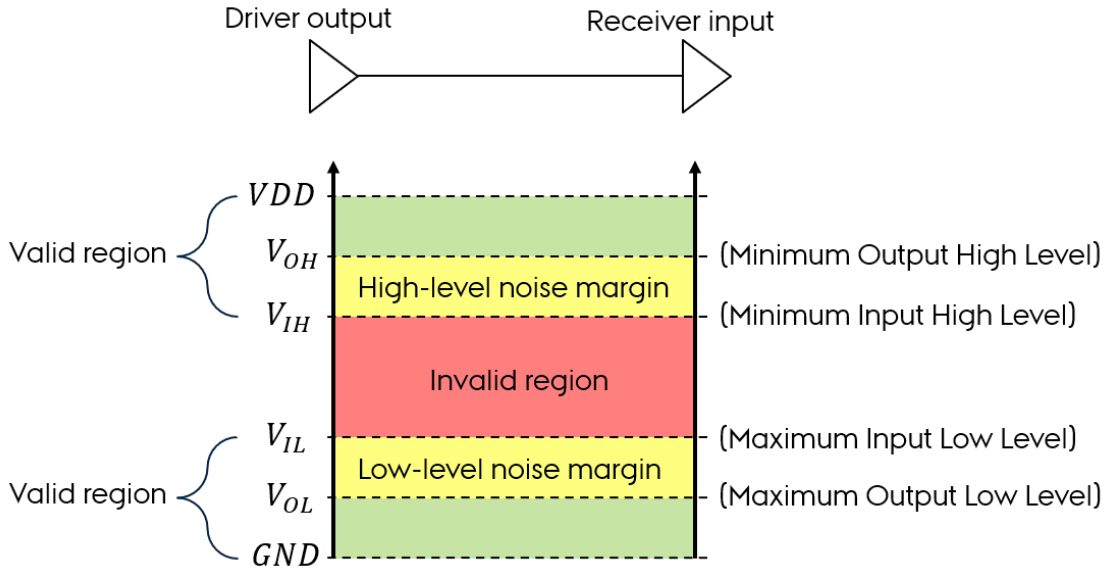


Figure 4. Illustration of high- and low-level noise margin between driver output and receiver input expected voltage levels.

Part 2 – Quantitative Formulas vs Simulations

With equivalent circuit models designed in part 1 for each of the 3 cases, we now put them to use to predict the circuit response and find the crosstalk on the quiet line (victim line) as a response to driving a clock signal on the driven line (aggressor line).

Note, the quiet line on the board is locked to low level through a resistor. However, noise is noise and would be present on a high-level quiet line as well as low level. The only difference is whether it is the positive or negative noise spikes that might cause the quiet line to violate the input limits and force it into the invalid region (see Figure 4).

Task 2.1 Quantitative results

For each transmission line case use the quantitative formulas from the book (HSDSD[2]) page 48 figure 3.4 and determine the crosstalk on both the far (receiver) and near (driver) end of the quiet line. Note that neither the aggressor nor victim line are matched in the far end.

(Hint: None of the cases in figure 3.4 matches exactly, however, the case on the board can be achieved by cleverly combining two of the other cases in the book by use of superposition.)

Based on the found victim line crosstalk for each transmission line case, determine for both the near end (NEXT) and far end (FEXT):

- The peak positive going noise level.
- The peak negative going noise level.
- High level remaining noise margin, or lack thereof.
- Low level remaining noise margin, or lack thereof.

Is it acceptable (discuss)?

Task 2.2 Simulation results

For each transmission line case set up a simulation (LTspice) and observe the driver and quiet line at both ends (use probes). Ensure to set a simulation time long enough to see the full effect of the reflections on the lines. (Reminder, edge rate of $500\ ps$, clock source of $20\ MHz$, $3.3\ V$ logic.)

Document the voltage curves over time, at the driver and receiver ends of both the aggressor line and the crosstalk on the victim line.

Based on the found victim line crosstalk for each transmission line case, determine for both the near end (NEXT) and far end (FEXT):

- The peak positive going noise level.
- The peak negative going noise level.
- High level remaining noise margin, or lack thereof.
- Low level remaining noise margin, or lack thereof.

Is it acceptable (discuss)?

[2] High-Speed Digital System Design

Task 2.3

Compare the results of the quantitative model versus the simulation model. How do they compare?

Were the noise levels larger than the noise margins?

Consider here the actual signal levels versus the receivers acceptable input range, high and low levels. Did the signals stay within the valid ranges?

Remember to consider the worst-case scenarios.

Part 3 - Measurements

This part is all done in the laboratory using the HW test board and an RTO 1024 oscilloscope to measure the signals.

Initial setup

Connect the board to the oscilloscope via the USB ports to power the board.

Adjust the supply voltage to $3.3\ VDC$, measuring it using the VCC probe point, P45, on the board. (Use a multimeter for measuring this. If using an oscilloscope use a standard high-impedance probe for measuring this.)

Probe setup

Connect the custom-made coaxial probes to scope channels 1 and 2. (Probes are found in the laboratory together with the boards.)

Afterwards, set both channels to $50\ \Omega$ input mode, and the channels attenuation for $21x$.
For measurement with $1000\ \Omega$ series resistor and $50\ \Omega$ scope input, we have: $\frac{1050\ \Omega}{50\ \Omega} = 21$
Why is this required? Explain. What are we correcting for here?

Board setup

For each measurement, remember to only enable the output(s) that you need using the jumpers. (Again, study the schematic for details.)

For each measurement, trigger the oscilloscope with the aggressor signal corresponding to the victim line that you are measuring on for timing reference.

Place the probes with the correct polarity, else you may risk overloading the oscilloscope USB output and/or damage the probe channel in the oscilloscope! THIS IS SERIOUS! PAY ATTENTION!!!

Task 3.1 Measurement results, nominal supply

For each transmission line case measure and document the crosstalk on the victim line at both far and near end. Remember to keep the aggressor signal in the plots for reference.

Based on the found victim line crosstalk for each transmission line case, determine for both the near end (NEXT) and far end (FEXT):

- The peak positive going noise level.
- The peak negative going noise level.
- High level remaining noise margin, or lack thereof.
- Low level remaining noise margin, or lack thereof.

Is it acceptable (discuss)?

Task 3.2 Measurement results, reduced supply

Reduce the supply voltage by 25% and repeat the measurements in task 3.1.

How does the reduced voltage supply affect the crosstalk and noise levels?

Is it acceptable (discuss)?

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About the report

The final report for Laboratory Assignment 1 - Crosstalk must include answers to all the tasks and questions above.

Also include a section where the quantitative, simulation and measurement results are compared and discussed, including a discussion/analysis of potential reasons for mismatch and other sources of errors.

If time permits, try to adjust the models to better match the realities of the measurement results.

When writing the report please be short, be precise, but do explain what you do/see/measure/observe/simulate/etc. Remember to document results so the reader can see the source of the results.

Please do not use generative AI for writing text but use your own words. The purpose is clarifying your own understanding, so that you yourself understand what is happening. There are no minimum or maximum requirements for the length, but remember, this is a laboratory report, not a master thesis!

The report is handed in via a BrightSpace group assignment, where you can also see the deadline for it. One report per group, as a single PDF file.